



Q1. I am working on a project that is based on one IR sensor connected to PIC 18F4550. I need to connect 16 IR sensors to the microcontroller. Is it possible to do that or else, how many IRs can be connected at a time?

Vinothkumar
Tuticorin, India

A1. You will not be able to use PIC 18F4550 microcontroller for your project as it has only 13 analogue input channels (five on port A, five on port B and three on port E). Instead, you can use PIC16F1713 for your project. This microcontroller has a 10-bit analogue-to-digital converter (ADC), built-in module with a handling capacity of 17 analogue input channels (five on port A, six on port B and six on port C). If you are familiar with the programming of PIC 18F4550, you will not find it difficult to work with PIC16F1713. That is the advantage of working with PIC microcontrollers as these have common IDEs and migration within the family is easy.

Q2. I want to know more about LED dot-matrix in a simple circuit.

Mohd Waseem Ansari
Through E-mail

A2. A simple project using three dot-matrix for display of a scrolling message has been published in EFY November 2014 issue.

Q3. Please provide information on controlling the speed of a DC motor using 8051.

Ruchita Shinde
Through E-mail

A3. The speed of a DC motor is controlled by running it with varying DC voltage, which we can get by configuring the microcontroller for PWM operation. However, 8051 microcontroller does not have an internal PWM peripheral. This mode is achieved by having a pulse train with different

mark and space timings, that is, different duty cycles by using different delay routines. The microcontroller is a low-power device, hence, will not be able to meet the current requirement of the motor, and a suitable driver circuit would be required. Most popular driver IC is L293D, which has two H bridges, so you can control the direction and braking of two DC motors. Interfacing this driver IC also provides protection to the microcontroller against voltage spikes produced while reversing the direction of the motor. The solution is to use a motor driver circuit in between the microcontroller and DC motor. A very good circuit using AT89C51 microcontroller has been published in June 2009 issue of EFY. You could refer the same for your project.

Q4. I have the following doubts:

1. What are the advantages of using a JK flip-flop in circuit?

2. Is negative-edge triggering better than positive-edge triggering?

3. What is a re-triggerable monostable multivibrator?

Meenakshi Sain
Pilani

A4. The JK flip-flop is the most versatile and widely-used of all designs. It is considered to be a universal flip-flop. An SR flip-flop can be replaced by a JK flip-flop without any change in operation. However, you cannot replace a JK flip-flop with an SR flip-flop, as S=1 and R=1 condition is not allowed, but J=1 and K=1 condition is permitted. The JK flip-flop has no invalid or forbidden input states of the SR latch. It can also act as a T flip-flop to accomplish a toggling action if J and K are tied together. This mode finds extensive use in binary counters.

An edge-triggered flip-flop changes states either at positive-edge (rising-edge) or negative-edge (falling-edge) of the clock pulse on the control input. The data lockout occurs at the end of the edge. The multi-transition problem can be stopped with positive- or negative-edge triggering. Both types are used as synchronous counters count on the rising-edge and ripple counters count on the falling edge. IC SN 74111

and IC 74112 are positive-level triggered and negative-level triggered flip-flops, respectively.

A monostable multivibrator has one stable state with zero output at pin 3. When triggered by input pulse at pin 2, the multivibrator switches to its unstable position, high at pin 3 for a period of time and then returns to its stable state. The time period it remains in unstable state is given by

$$t = \ln(2)R_1C_1$$

where R_1 is connected between pin 6 and pin 7, shorted and power supply. C_1 is connected between pin 6 and pin 7, shorted and ground. If repeated application of the input pulse keeps the circuit in an unstable state, it is called a retriggerable monostable. If further trigger pulses do not affect the period, the circuit is a non-retriggerable multivibrator.

Q5. In many circuits and PCBs, there is a six-terminal on/off switch mentioned. What is it called, exactly? If possible, attach a picture.

Upamanyu Das
Kolkata

A5. A double-pole double-throw switch (DPDT) will have six terminals. Pole refers to the number of circuits controlled by the switch and throw refers to the extreme position of the actuator. A DPDT switch functions like two separate single-pole double throw (SPDT) switches, operated by the same actuator.

The arrangement of poles and throws in a DPDT switch is shown in Fig. 1. Here, pin 3 and pin 4 are poles,

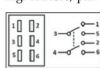


Fig. 1: DPDT switch

pin 1 and pin 2 are a set of throws, and pin 5 and pin 6 is another set of throws. In position one of the actuator, pin 3 is connected to pin 1 and pin 4 is connected to pin 2. In the second position of the actuator, pin 3 is connected to pin 5 and pin 4 is connected to pin 6. The operation can be slider-type or toggle-type.

Answers compiled by EFY joint director (training), Col. N.C. Pande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender